

Siqi Yang

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Research Interests

Neuromorphic Imaging (Spike Camera), Inverse Rendering & Relighting, Video Generation & World Models

Education

- | | |
|-------------------|--|
| 09/2022 - Present | Peking University , Beijing, China
<i>Ph.D. candidate in Computer Science, Camera Intelligence Lab</i>
<i>Advised by Prof. Boxin Shi</i>
<i>Expected graduation: July 2027</i> |
| 09/2018 - 07/2022 | Peking University , Beijing, China
<i>B.S. in Computer Science, Turing Honor Class</i> |

Experience

- | | |
|-------------------|--|
| 12/2025 - Present | Shanda AI Research Tokyo , Tokyo, Japan
<i>Research Intern</i>
Multi-player world models, panorama video generation, generative game platform |
| 05/2019 - 08/2019 | Hyperplane Lab, Peking University , Beijing, China
<i>Research Intern</i>
Convolutional neural network quantization and pruning |
| 07/2018 - 08/2018 | Youtu X-Lab, Tencent , Shenzhen, China
<i>Research Intern</i>
Object detection and classification on the Tencent Weishi dataset |

Awards and Honors

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| 2025 | | PKU-Huawei Turing Graduate Program Scholarship |
| 2025 | | Huawei Hackathon, Software Challenge , Gold Medal |
| 2019 | | Peking University Academic Excellence Award |
| 2018 | | China Collegiate Programming Contest (CCPC), Guilin , Gold Medal |
| 2017 | | National Olympiad in Informatics (NOI) , Silver Medal |
| 2016 – 2017 | | Other Informatics Olympiads , NOIP First Prize (×2); APIO, CTSC, WC Bronze Medals |

Academic Service

Conference Reviewer: CVPR · ICCV · ECCV · NeurIPS · SIGGRAPH · ACCV · PRCV · ICIG

Journal Reviewer: TPAMI

Teaching

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|-------------|---|
| 2021 & 2022 | Algorithm Design and Analysis
<i>Teaching Assistant, Peking University</i> |
| 2020 | Introduction to Computing
<i>Teaching Assistant, Peking University</i> |
| 2017 - 2020 | Informatics Olympiad (NOI/NOIP) Training
<i>Guest Lecturer, training camps in Beijing, Jinan, Qingdao, and Shijiazhuang</i> |

Publications

* denotes equal contribution.

- Ziqi Cai, **Siqi Yang**, Yimu Wang, Zixian Gao, Yunheng Liu, Shuchen Weng, Erwin Wu, Kaipeng Zhang, Boxin Shi. [MASS: Multiplayer World Models with Authoritative Shared State](#). *arXiv preprint*, 2026.
- Guixu Lin, Zheng-Hui Huang, **Siqi Yang**, Ming-Hsuan Yang, Kaipeng Zhang, Zhixiang Wang. [Generative World Renderer at the Speed of Play](#). *arXiv preprint*, 2026.
- Chu Zhou*, **Siqi Yang***, Kailong Zhang, Heng Guo, Zhaofei Yu, Boxin Shi, Imari Sato. [High-Speed Full-Color HDR Imaging via Unwrapping Modulo-Encoded Spike Streams](#). *arXiv preprint*, 2026.
- Haojie Zheng, Yixin Yang, **Siqi Yang**, Shuchen Weng, Boxin Shi. [InstructAV2AV: Instruction-Guided Audio-Video Joint Editing](#). *SIGGRAPH Asia*, 2026 (to appear).
- Jiaming Tan, Zhen Li, Shuwei Shi, Minggui Teng, **Siqi Yang**, Yuwei Wu, Bo Zheng, Kaipeng Zhang, Chuanhao Li. Interactive Panoramic World Exploration via Camera Control. *SIGGRAPH Asia*, 2026 (to appear).
- Yakun Chang, Zhaojun Huang, **Siqi Yang**, Yeliduosi Xiaokaiti, Shikui Wei, Yao Zhao, Tiejun Huang, Boxin Shi. [HFR and HDR Video from Multi-Attenuated Spikes Using a Rapidly Rotating SpokeND Filter](#). *CVPR*, 2026.
- Haojie Zheng, Shuchen Weng, Jingqi Liu, **Siqi Yang**, Boxin Shi, Xinlong Wang. [Audio-sync Video Instance Editing with Granularity-Aware Mask Refiner](#). *CVPR*, 2026.
- Yifei Xia, Shuchen Weng, **Siqi Yang**, Jingqi Liu, Chengxuan Zhu, Minggui Teng, Zijian Jia, Han Jiang, Boxin Shi. [PanoWan: Lifting Diffusion Video Generation Models to 360° with Latitude/Longitude-aware Mechanisms](#). *NeurIPS*, 2025.
- **Siqi Yang**, Jinxiu Liang, Zhaojun Huang, Yeliduosi Xiaokaiti, Yakun Chang, Zhaofei Yu, Boxin Shi. [SpikeDiff: Zero-shot High-Quality Video Reconstruction from Chromatic Spike Camera and Sub-Millisecond Spike Streams](#). *ICCV*, 2025.
- Jinxiu Liang, Bohan Yu, **Siqi Yang**, Haotian Zhuang, Jieji Ren, Peiqi Duan, Boxin Shi. [EventUPS: Uncalibrated Photometric Stereo Using an Event Camera](#). *ICCV*, 2025.
- **Siqi Yang***, Zhaojun Huang*, Yakun Chang, Bin Fan, Zhaofei Yu, Boxin Shi. [Real-Data-Driven 2000 FPS Color Video from Mosaicked Chromatic Spikes](#). *ECCV*, 2024.
- **Siqi Yang**. Focal Stack and Light Field Imaging (Chapter 5). In [Computational Photography: Imaging Model Theory and Deep Learning Practice](#), Boxin Shi (Ed.), *China Machine Press*, 2024.
- **Siqi Yang***, Xuanning Cui*, Yongjie Zhu, Jiajun Tang, Si Li, Zhaofei Yu, Boxin Shi. [Complementary Intrinsic From Neural Radiance Fields and CNNs for Outdoor Scene Relighting](#). *CVPR*, 2023.
- Bohan Yu, **Siqi Yang**, Xuanning Cui, Siyan Dong, Baoquan Chen, Boxin Shi. [MILO: Multi-bounce Inverse Rendering for Indoor Scene with Light-emitting Objects](#). *TPAMI*, 2023.

Patents

- Method and Apparatus for Self-Calibrating Surface Normal Field Reconstruction Based on Asynchronous Event Streams. China patent application, pending.

Skills

Programming: Python · C++ · CUDA · Rust · Lisp · Nix

Deep Learning: PyTorch · JAX · TensorFlow (v1 & v2) · TensorLayer · Keras

Graphics & Imaging: NeRF · 3DGS · Spike sensor pipeline · Differentiable rendering

Infrastructure: Docker · Ray · Cloudflare Workers · GCP · Git

Languages: Chinese · English